MODEL SCOPE

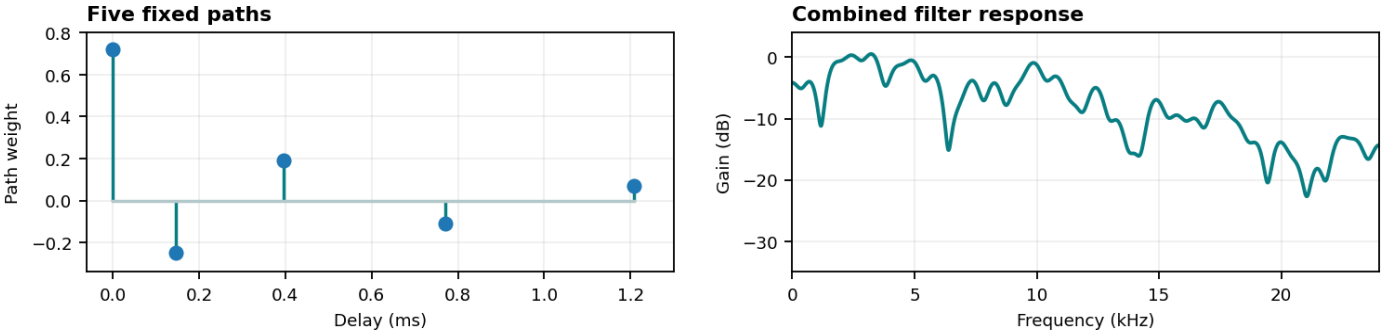
What the acoustic room simulator simulates

The OFDM text dashboard's "Colored-room simulation" routes generated audio through a deterministic software channel. It tests whether the modem and transfer protocol recover data after simple echo, frequency coloring, noise, and start-time uncertainty. It does not calculate the acoustics of a particular room. [1-3]

In one line: 48 ksample/s OFDM waveform -> five-path filter -> three-tap spectral tilt -> seeded Gaussian noise -> decoder and integrity checks.

The channel itself

Effect	Implementation	Meaning
Multipath / echo	Five FIR taps at 0, 7, 19, 37, 58 samples; weights +0.72, -0.25, +0.19, -0.11, +0.07.	Copies of the signal arrive up to 1.208 ms after the direct path. Negative weights represent phase inversion.
Spectral color	Convolution with [0.20, 0.60, 0.20].	A small smoothing filter stands in for speaker / microphone roll-off; it also changes the effective impulse response.
Noise	Zero-mean Gaussian samples with standard deviation 10^(noise_dbfs / 20); fixed RNG seed.	Dashboard default is -60 dBFS. Noise is repeatable and white; "colored" describes the filtered signal, not the noise.
Start offset	733 zero samples before a batch; 613 + (sequence x 17) mod 193 for single packets.	Tests receiver acquisition with about 13-17 ms of extra leading silence; this is not a distance model.



Fixed filter response calculated from the source taps. At 48 ksample/s, the final effective tap is at sample 60 (1.25 ms), within the default 64-sample cyclic prefix (1.33 ms). [1, 4]

END-TO-END BEHAVIOR

What a successful dashboard run proves

The room model is one stage in a larger software loopback. The default text service sends UTF-8 chunks through the real modem encoder and decoder, then validates the recovered bytes. [2-4]

Stage	Exercised in simulation
Transmit	Text framing and sequence numbers; 48 ksample/s real-valued OFDM; selectable QPSK or experimental 8-PSK, FFT size, band, and coding. The current dashboard defaults to QPSK, FFT-256, v3 rate-2/3, and a requested 2-15 kHz band (2.0625-15 kHz on the bin grid).
Receive	Start detection, three training symbols for channel estimation, pilot-based phase / slope correction, body decoding, packet CRC and outer text-frame CRC.
Transfer	A batch uses a continuous superframe of up to 64 slots. Failed packets can be retried individually (two retries by default). Valid chunks appear in order; final output SHA-256 must match the source.
Display	The waterfall is derived from the same simulated received samples, converted to 16-bit PCM for the monitor. Its spectrum is a software result, not a microphone measurement.

What the model does not establish

No physical room geometry: no room dimensions, wall materials, reflection paths calculated from position, distance-dependent attenuation, or measured impulse response. The tap delays and gains are hard-coded. [1]

No moving or imperfect hardware path: the filter is time-invariant; this backend adds no Doppler, carrier frequency offset, independent audio-clock drift, nonlinear speaker behavior, clipping, automatic gain control, or real ambient-noise recording. Decoder correction features may exist, but this channel does not inject those effects. [1-3]

No over-air or FPGA evidence: loopback does not play sound or use the FPGA AXIS path. Those are separate dashboard backends. The configured -60 dBFS noise level is a digital sample setting, not a measured acoustic SNR or SPL. [2, 3]

Verification and interpretation

A local smoke check on this workspace encoded one v3 rate-2/3 packet at the dashboard's 2-15 kHz setting, applied the seeded -60 dBFS channel and 733-sample offset, and decoded the exact payload with a valid result. This confirms one deterministic test case, not a field reliability rate.

Use this mode to demonstrate packet processing and recovery from a modest, repeatable channel. Use Speaker-to-lapel-microphone mode to measure the actual audio path; compare packet error rate and received SNR separately.

Source trail

[1] software/cora-ofdm-acoustic/cora_ofdm_acoustic.py, lines 31-44, 1082-1116, 1549-1563.

[2] software/cora-ofdm-text-demo/cora_ofdm_text_transfer.py, lines 113-182, 928-1215.

[3] software/cora-ofdm-text-demo/cora_ofdm_text_dashboard.py, lines 105-175, 343-370, 609-642.

[4] software/cora-ofdm-text-demo/README.md, lines 3-42; software/cora-ofdm-acoustic/cora_ofdm_acoustic.py, lines 924-1022.